

```
exercise1.py x
1 class CelestialBody:
2     def __init__(self, size, mass, composition, name):
3         self.size = size
4         self.mass = mass
5         self.composition = composition
6         self.name = name
7
8 # create the satellite class
9 class Satellite(CelestialBody):
10     def __init__(self, size, mass, composition, name, host_planet):
11         super().__init__(size, mass, composition, name)
12         self.host_planet = host_planet
13
14 # create the planet class
15 class Planet(CelestialBody):
16     def __init__(self, size, mass, composition, name, host_star):
17         super().__init__(size, mass, composition, name)
18         self.host_star = host_star
19
20
```

Inheritance Exercise 1

Exercise 1

Use the `CelestialBody` class to the left as the parent class. Create the `Satellite` and `Planet` classes, both of which are children of `CelestialBody`. Using the `super()` keyword to extend the `__init__` method of the parent class, add the attribute `host_planet` to the `Satellite` class and add the attribute `host_star` to the `Planet` class.

Expected Attributes

After extending the `__init__` method, your classes should have the following attributes.

CelestialBody	Satellite	Planet
size	size	size
mass	mass	mass
composition	composition	composition
name	name	name
	host_planet	host_star

TRY IT

Command was successfully executed.

Submit your code when ready.

Here is one possible solution:

```
class CelestialBody:
    def __init__(self, size, mass, composition, name):
        self.size = size
        self.mass = mass
        self.composition = composition
        self.name = name

# create the satellite class
class Satellite(CelestialBody):
    def __init__(self, size, mass, composition, name, host_planet):
        super().__init__(size, mass, composition, name)
        self.host_planet = host_planet

# create the planet class
class Planet(CelestialBody):
    def __init__(self, size, mass, composition, name, host_star):
        super().__init__(size, mass, composition, name)
        self.host_star = host_star
```

Check It!

LAST RUN on 12/28/2023, 12:50:24 AM

Test passed

Next